

Design and Implementation of an AI-Based Customer Support Chatbot Using Natural Language Processing

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Course: AI & ML

Abstract

This project focuses on designing an AI-based chatbot using NLP and Machine Learning. It classifies user queries into intents like order tracking and refunds using deep learning and transformer models. The system improves automation and customer experience.

Acknowledgement

I sincerely thank my faculty, mentors, institution, family, and friends for their support and guidance.

Introduction

Handling customer queries manually is inefficient. AI chatbots automate interactions using NLP and ML, providing faster and consistent responses.

The main goal of this project is to demonstrate how AI and NLP can be effectively applied to automate customer support services, reduce human effort, and improve efficiency. This system provides a foundation for developing more advanced conversational agents that can be integrated into real world applications

Literature Review

AI-based chatbots have become an essential part of modern customer service systems. Traditional rule based chatbots were limited in handling complex queries. With the advancement of Natural Language Processing (NLP) and deep learning, intelligent chatbots can now understand user intent and provide accurate responses.

Recent research highlights the use of transformer – based models such as BERT and GPT for improving conversational accuracy. These models helps in understanding context and generating human like responses.

This project utilizes similar NLP techniques to build an efficient chatbot sytem

Objectives & Scope

Objectives:

- Build chatbot
 - Classify intents
- NLP preprocessing
- Deploy web app

Scope:

- E-commerce & support
- Reduce manual work
- Extendable features

Problem Statement

Customer support systems face delays and high costs. This project creates an automated chatbot for instant responses.

Methodology

Data collected in JSON. NLP preprocessing using NLTK. Model trained with TensorFlow. Transformer models for responses. Flask for deployment.

Enhanced Methodology

Data Collection

- Dataset prepared in JSON format containing intents, patterns, and responses

Text Preprocessing

- Tokenization
- Stopword removal
- Lemmatization using NLTK

Feature Extraction

- Bag of words / TF-IDF
- Words embeddings for better context understanding

Model Training

- Deep learning model built using TensorFlow
- Classification of user queries into predefined intents

Response Generation

- Matching predicted intent with predefined responses
- Transformer based improvement for dynamic replies

Deployment

- Web application built using Flask

Applications

the chatbot systems can be used in multiple domain such as:

- E-commerce customer support
- Banking and financial services
- Healthcare assistance
- Educational platforms
- Technical support systems

It helps in organization reduces workload and improve user satisfaction.

Limitations

Limitations:

- Limited dataset may reduce accuracy
- Cannot handle highly complex queries
- Depends on predefined intents
- No real time learning capability

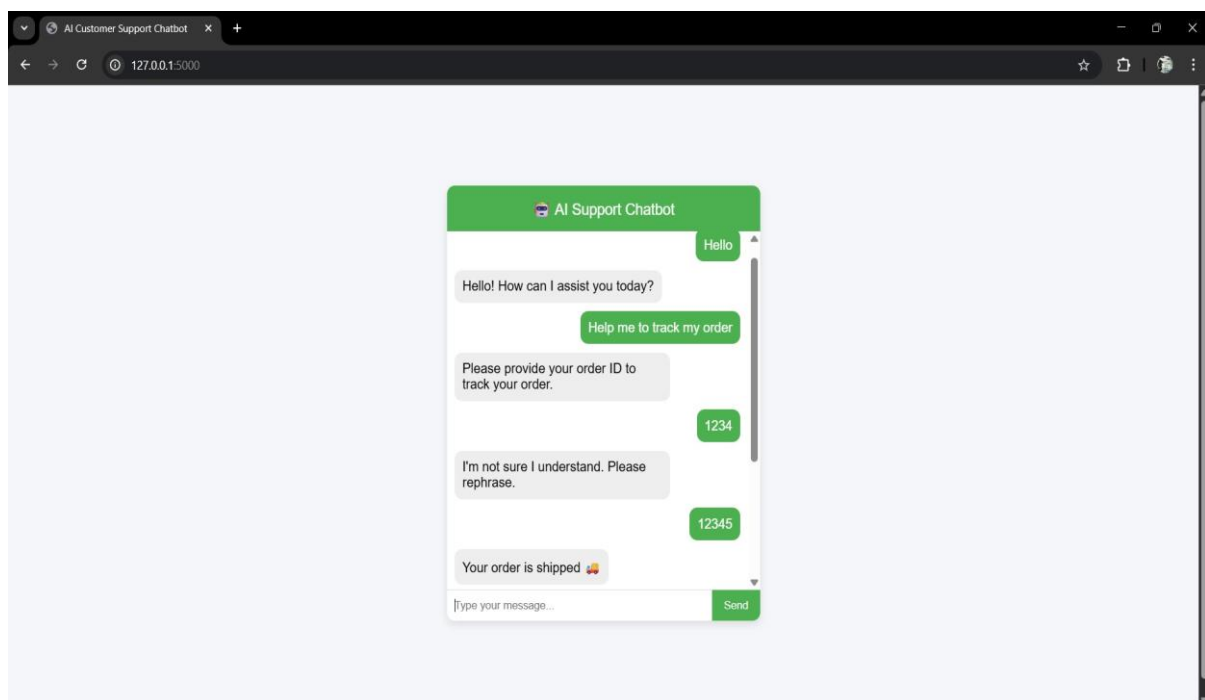
Implementation

```

1 print("STARTING TRAINING...")
2 import nltk
3 nltk.download('punkt_tab')
4 import json
5 import nltk
6 import numpy as np
7 from nltk.stem import WordNetLemmatizer
8 from sklearn.preprocessing import LabelEncoder
9 from tensorflow.keras.models import Sequential
10 from tensorflow.keras.layers import Dense, Dropout
11
12 nltk.download('punkt')
13 nltk.download('wordnet')
14 nltk.download(['punkt_tab'])
15
16 lemmatizer = WordNetLemmatizer()
17
18 import os
19
20 BASE_DIR = os.path.dirname(os.path.abspath(__file__))
21 file_path = os.path.join(BASE_DIR, 'intents.json')
22
23 with open(r"C:\Users\sreeh\OneDrive\Desktop\INTERNSHIP PROJECT\chatbot_project\intents.json") as f:

```

The terminal output shows the chatbot running on a local server (127.0.0.1) and handling several HTTP requests. The status bar indicates the file is at line 14, column 27, using Python 3.13.7.



Results

The chatbot accurately responds to queries and maintains context through a web interface.

Conclusion

The chatbot demonstrates effective use of AI and NLP, improving response time and reducing effort.

This project demonstrates the successful design and implementation of AI – based chatbot using NLP and machine learning techniques. the chatbot effectively classifies user queries and provides accurate responses. The system enhances customer support by reducing manual effort and improving efficiency. With further improvements, it can be deployed in real world applications

Future Scope

- Transformers
- Voice support
- Multilingual
- Cloud deployment

References

Python Docs

TensorFlow Docs

NLTK Docs

ML Research